Quizrr Chapter-Wise Test for JEE Main - 2025

Q.1

If $OABC$ is a tetrahedron where O is the origin and A, B, C are three other vertices with position vectors $\vec{a}, \vec{b}$ and $\vec{c}$ respectively, then the centre of sphere circumscribing the tetrahedron is given by the position vector

(A) $\frac{a^2(\vec{b}\times\vec{c})+b^2(\vec{c}\times\vec{a})+c^2(\vec{a}\times\vec{b})}{2[\vec{a}\vec{b}\vec{c}]}$

(B) $\frac{b^2(\vec{b}\times\vec{c})+a^2(\vec{c}\times\vec{a})+c^2(\vec{a}\times\vec{b})}{[\vec{a}\vec{b}\vec{c}]}$

(C) $\frac{b^2(\vec{b}\times\vec{c})+a^2(\vec{c}\times\vec{a})+c^2(\vec{a}\times\vec{b})}{2[\vec{a}\vec{b}\vec{c}]}$

(D) $\frac{a^2(\vec{a}\times\vec{b})+b^2(\vec{b}\times\vec{c})+c^2(\vec{c}\times\vec{a})}{2[\vec{a}\vec{b}\vec{c}]}$

Q.2

Let O be an interior point of ΔABC, such that $\overrightarrow{OA}+2\overrightarrow{OB}+3\overrightarrow{OC}=\vec{0}$. Then the ratio of the area of ΔABC to the area of ΔAOC is

(A) 2

(B) $\frac{3}{2}$

(C) 3

(D) $\frac{5}{2}$

Q.3

Let $A4,7,8,\ B2,3,4$ and $C2,5,7$ be the position vectors of the vertices of a ΔABC. The length of the internal bisector of the angle of A is

(A) $\frac{3}{2}\sqrt{34}$

(B) $\frac{2}{3}\sqrt{34}$

(C) $\frac{1}{2}\sqrt{34}$

(D) $\frac{1}{3}\sqrt{34}$

Q.4

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If $\vec{a}, \vec{b} \& \vec{c}$ are three non-coplanar non-zero vectors, then $(\vec{a} \cdot \vec{a})(\vec{b} \times \vec{c}) + (\vec{a} \cdot \vec{b})(\vec{c} \times \vec{a}) + (\vec{a} \cdot \vec{c})(\vec{a} \times \vec{b})$ is equal to

(A) $[\vec{b}\vec{c}\vec{a}]\vec{a}$

(B) $[\vec{c}\vec{a}\vec{b}]\vec{b}$

(C) $[\vec{a}\vec{b}\vec{c}]\vec{c}$

(D) None of these

Q.5

If $\vec{b}$ and $\vec{c}$ are any two non-collinear mutually perpendicular unit vectors and $\vec{a}$ is any vector, then $(\vec{a} \cdot \vec{b})\vec{b} + (\vec{a} \cdot \vec{c})\vec{c} + \frac{\vec{a} \cdot (\vec{b} \times \vec{c})}{|\vec{b} \times \vec{c}|^2}(\vec{b} \times \vec{c})$ is equal to :

(A) $\overrightarrow{a}$

(B) $2\overrightarrow{a}$

(C) $3\overrightarrow{a}$

(D) None

Q.6

If $4\hat{i} + 7\hat{j} + 8\hat{k}, 2\hat{i} + 3\hat{j} + 4\hat{k}$ and $2\hat{i} + 5\hat{j} + 7\hat{k}$ are the position vectors of the vertices A, B and C respectively of triangle ABC. The position vector of the point where the bisector of angle A meets BC is:

(A) $\frac{1}{3}(6\hat{i} + 13\hat{j} + 18\hat{k})$

(B) $\frac{2}{3}(6\hat{i} + 12\hat{j} - 8\hat{k})$

(C) $\frac{1}{3}(-6\hat{i} - 8\hat{j} - 9\hat{k})$

(D) $\frac{2}{3}(-6\hat{i} - 12\hat{j} + 8\hat{k})$

Q.7

The vector $\vec{a} = \alpha\hat{i} + 2\hat{j} + \beta\hat{k}$ lies in the plane of the vectors $\vec{b} = \hat{i} + \hat{j}$ and $\vec{c} = \hat{j} + \hat{k}$ and bisects the angle between $\vec{b}$ and $\vec{c}$. Then which one of the following gives possible values of α and β?

(A) $\alpha = 2, \beta = 2$

(B) $\alpha = 1, \beta = 2$

(C) $\alpha = 2, \beta = 1$

(D) $\alpha = 1, \beta = 1$

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Q.8

If $\vec{A} = \hat{i} + 2\hat{j} + 3\hat{k}$, $\vec{B} = -\hat{i} + 2\hat{j} + \hat{k}$ and $\vec{C} = 3\hat{i} + \hat{j}$, evaluate t, if the vector $\vec{A} + t\vec{B}$ and $\vec{C}$ are mutually perpendicular.

(A) 5

(B) 4

(C) 1

(D) 2

Q.9

If a and $\vec{b}$ are non-collinear vectors, then the value of α for which the vectors $\vec{u} = (\alpha - 2)\vec{a} + \vec{b}$ and $\vec{v} = (2 + 3\alpha)\vec{a} - 3\vec{b}$
are collinear is :

(A) $\frac{3}{2}$

(B) $\frac{2}{3}$

(C) $-\frac{3}{2}$

(D) $-\frac{2}{3}$

Q.10

The points $\mathrm{D}, \mathrm{E}, \mathrm{F}$ divide BC, CA and AB of the triangle ABC in the ratio $1:4, 3:2$ and $3:7$ respectively and the point K divides AB in the ratio $1:3$, then $(\overrightarrow{\mathrm{AD}} + \overrightarrow{\mathrm{BE}} + \overrightarrow{\mathrm{CF}}) : \overrightarrow{\mathrm{CK}}$ is
equal to

(A) $1:1$

(B) $2:5$

(C) $5:2$

(D) None of these

Q.11

$ABCD$ a parallelogram, and A_1 and B_1 are the midpoints of sides BC and CD, respectively. If $\overrightarrow{AA_1} + \overrightarrow{AB_1} = \lambda\overrightarrow{AC}$, then λ is equal to

(A) $\frac{1}{2}$

(B) 1

(C) $\frac{3}{2}$

(D) 2

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Q.12

The vector $\vec{c}$ directed along the bisectors of
the angle between the vectors $\vec{a} = 7\hat{i} - 4\hat{j} - 4\hat{k}$,
$\vec{b} = -2\hat{i} - \hat{j} + 2\hat{k}$ and $|\vec{c}| = 3\sqrt{6}$ is given by

(A) $\hat{i} - 7\hat{j} + 2\hat{k}$

(B) $\hat{i} + 7\hat{j} - 2\hat{k}$

(C) $\hat{i} + 7\hat{j} + 2\hat{k}$

(D) $\hat{i} + 7\hat{j} + 3\hat{k}$

Q.13

If the vectors $\alpha\hat{i} + \alpha\hat{j} + \gamma\hat{k}, \hat{i} + \hat{k}$ and $\gamma\hat{i} + \gamma\hat{j} + \beta\hat{k}$
lie on a plane, where α, β and γ are distinct nonnegative numbers, then γ is

(A) Arithmetic mean of α and β

(B) Geometric mean of α and β

(C) Harmonic mean of α and β

(D) None of the above

Q.14

Let $\vec{a}, \vec{b}$ and $\vec{c}$ be three non-zero vectors such that no two of these are collinear. If the vector $\vec{a} + 2\vec{b}$ is collinear with $\vec{c}$ and $\vec{b} + 3\vec{c}$ is collinear with $\vec{a}$ (\lambda being some non-zero scalar) then $\vec{a} + 2\vec{b} + 6\vec{c}$ equals

(A) 0

(B) $\lambda\vec{b}$

(C) $\lambda\vec{c}$

(D) $\lambda\vec{a}$

Q.15

Let α, β and y be distinct real numbers. The points with position vectors $\alpha, i + \beta j + \gamma k, \beta i + \gamma j + \alpha k$ and $\gamma i + \alpha j + \beta k$

(A) are collinear

(B) form an equilateral triangle

(C) form a scalene triangle

(D) form a right-angled triangle

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Q.16

If $\vec{a} = \hat{i} + \hat{j} + \hat{k}, \vec{b} = 4\hat{i} + 3\hat{j} + 4\hat{k}$
and $\vec{c} = \hat{i} + \alpha\hat{j} + \beta\hat{k}$ are coplanar and $|\vec{c}| = \sqrt{3}$, then

(A) $\alpha = \sqrt{2}, \beta = 1$

(B) $\alpha = 1, \beta = \pm 1$

(C) $\alpha = \pm 1, \beta = 1$

(D) $\alpha = \pm 1, \beta = -1$

Q.17

If $\vec{a}$ & $\vec{b}$ are two non-zero and non-collinear vectors, then $\vec{a} + \vec{b}$ and $\vec{a} - \vec{b}$ are

(A) linearly dependent vectors

(B) linearly independent vectors

(C) linearly dependent and independent vectors

(D) None of the above

Q.18

$ABCDEF$ is a regular hexagon where centre O is the origin. If the position vectors of A and B are $\hat{i} - \hat{j} + 2\hat{k}$ and $2\hat{i} + \hat{j} - \hat{k}$ respectively then $\overrightarrow{BC}$
is equal to

(A) $\hat{i} + \hat{j} - 2\hat{k}$

(B) $-\hat{i} + \hat{j} - 2\hat{k}$

(C) $3\hat{i} + 3\hat{j} - 4\hat{k}$

(D) None of these

Q.19

The vectors $\overrightarrow{AB} = 3\hat{i} + 5\hat{j} + 4\hat{k}$ and $\overrightarrow{AC} = 5\hat{i} - 5\hat{j} + 2\hat{k}$ are the sides of a triangle ABC. The length of the median through A is :

(A) $\sqrt{13}$ units

(B) $2\sqrt{5}$ units

(C) 5 units

(D) 10 units

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Q.20

If $\vec{a}, \vec{b}$ and $\vec{c}$ are the position vectors of the vertices of an equilateral triangle whose orthocentre is at the origin, then which one of the following is correct?

(A) $\vec{a}+\vec{b}+\vec{c}=\overrightarrow{0}$

(B) $\overrightarrow{\mathrm{a}}+\overrightarrow{\mathrm{b}}+\overrightarrow{\mathrm{c}}=$ unit vector

(C) $\vec{a}+\vec{b}=\vec{c}$

(D) $\overrightarrow{\mathrm{a}}=\overrightarrow{\mathrm{b}}+\overrightarrow{\mathrm{c}}$

Answers & Solutions

Q.1 Answer:

$\frac{a^2(\vec{b}\times\vec{c})+b^2(\vec{c}\times\vec{a})+c^2(\vec{a}\times\vec{b})}{2[\vec{a}\vec{b}\vec{c}]}$

Solution:

If the centre ' P ' is with position vector r,

then $\vec{a}-\vec{r}=\overrightarrow{PA}, \vec{b}-\vec{r}=\overrightarrow{PB}, \vec{c}-\vec{r}=\overrightarrow{PC}$

where $|\overrightarrow{PA}|=|\overrightarrow{PB}|$

$|\overrightarrow{PC}|=|\overrightarrow{OP}|=|\vec{r}|$

Consider $|\vec{a}-\vec{r}|=|\vec{r}|$

$\Rightarrow(\vec{a}-\vec{r})\cdot(\vec{a}-\vec{r})=\vec{r}\cdot\vec{r}$

$\Rightarrow a^2-2\vec{a}\cdot\vec{r}+r^2=r^2$

$\Rightarrow a^2=2\vec{a}\cdot\vec{r}$ **Similarly,** $b^2=2\vec{b}\cdot\vec{r}$ **and** $c^2=2\vec{c}\cdot\vec{r}$

Since, $(\vec{b}\times\vec{c}),(\vec{c}\times\vec{a})$ **and** $(\vec{a}\times\vec{b})$ **are non-coplanar, then** $\vec{r}=x(\vec{b}\times\vec{c})+y(\vec{c}\times\vec{a})+z(\vec{a}\times\vec{b})$

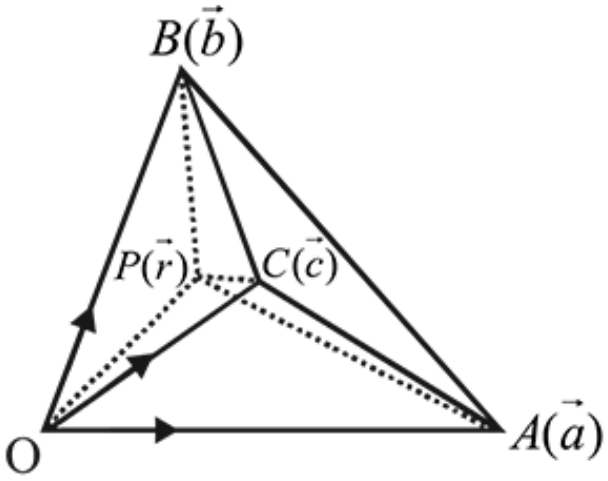
$\Rightarrow\vec{a}\cdot\vec{r}=x\vec{a}\cdot(\vec{b}\times\vec{c})+y\cdot0+z.0=x[\vec{a}\vec{b}\vec{c}]$

$\Rightarrow \quad x=\frac{\vec{a}\cdot\vec{r}}{[\vec{a}\vec{b}\vec{c}]}=\frac{a^2}{2[\vec{a}\vec{b}\vec{c}]}$

Similarly, $y=\frac{b^2}{2[\vec{a}\vec{b}\vec{c}]}$ **and** $z=\frac{c^2}{2[\vec{a}\vec{b}\vec{c}]}$

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Therefore, $\vec{r} = \frac{a^2(\vec{b}\times\vec{c})+b^2(\vec{c}\times\vec{a})+c^2(\vec{a}\times\vec{b})}{2[\vec{a}\vec{b}\vec{c}]}$



Q.2 Answer:

3

Solution:

Let O **be the origin. Then,**

$\vec{a} + 2\vec{b} + 3\vec{c} = \vec{0}$

$\Rightarrow \vec{a}\times\vec{b} = 3\vec{b}\times\vec{c}$ **(cross product with** $\vec{b}$**)**

$\Rightarrow \vec{a}\times\vec{c} = 2\vec{c}\times\vec{b}$ **(cross product with** $\vec{c}$**)**

Now,

$$\frac{\text{Area of } \Delta ABC}{\text{Area of } \Delta OCA} = \frac{\frac{1}{2}\overrightarrow{AB}\times\overrightarrow{AC}}{\frac{1}{2}\overrightarrow{OA}\times\overrightarrow{OC}}$$

$$\Rightarrow \frac{\text{Area of } \Delta ABC}{\text{Area of } \Delta OCA} = \frac{\left|\left(\vec{b}-\vec{a}\right)\times\left(\vec{c}-\vec{a}\right)\right|}{\left(\vec{a}\times\vec{c}\right)}$$

$$\Rightarrow \frac{\text{Area of } \Delta ABC}{\text{Area of } \Delta OCA} = \frac{\vec{b}\times\vec{c}-\vec{b}\times\vec{a}-\vec{a}\times\vec{c}}{\left|\vec{a}\times\vec{c}\right|}$$

$$\Rightarrow \frac{\text{Area of } \Delta ABC}{\text{Area of } \Delta OCA} = \frac{6\,|\,\vec{c}\times\vec{b}\,|}{2\,|\,\vec{c}\times\vec{b}\,|}$$

$$\Rightarrow \frac{\text{Area of } \Delta ABC}{\text{Area of } \Delta OCA} = 3$$

Q.3 Answer:

$\frac{2}{3}\sqrt{34}$

Solution:

The internal bisector $\overrightarrow{AB}$ **w.r.t.** $\overrightarrow{AB}$ **and** $\overrightarrow{AC}$ **is given by** $= \frac{\overrightarrow{AC}}{|\overrightarrow{AB}|+|\overrightarrow{AC}|}\overrightarrow{AB} + \frac{\overrightarrow{AB}}{|\overrightarrow{AB}|+|\overrightarrow{AC}|}\overrightarrow{AC}$

and $\overrightarrow{AB} = -2\hat{i} - \hat{j} - \hat{k} \Rightarrow |\overrightarrow{AB}| = 6$

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$\therefore \overrightarrow{AD} = \frac{1}{3}(-6i - 8i - 6k)$

$\Rightarrow \overrightarrow{AC} = \frac{2}{3}\sqrt{34}$

Q.4 Answer:

$[\vec{b}\vec{c}\vec{a}]\vec{a}$

Solution:

Since, $\vec{a}, \vec{b} \& \vec{c}$ **are non-coplanar**

$\therefore \vec{b} \times \vec{c}, \vec{c} \times \vec{a}, \vec{a} \times \vec{b}$ **are also non-coplanar.**

So, any vector can be expressed as a linear combination of these vectors.

Let, $\vec{r} = \lambda(\vec{b} \times \vec{c}) + \mu(\vec{c} \times \vec{a}) + \gamma(\vec{a} \times \vec{b})$

$\therefore \vec{a} \cdot \vec{r} = \lambda[\vec{a}\vec{b}\vec{c}] + \mu(0) + \gamma(0)1$

$\Rightarrow \lambda = \frac{\vec{a} \cdot \vec{r}}{[\vec{a}\vec{b}\vec{c}]}$

Similarly, $\mu = \frac{\vec{b} \cdot \vec{r}}{[\vec{a}\vec{b}\vec{c}]} \& \gamma = \frac{\vec{c} \cdot \vec{r}}{[\vec{a}\vec{b}\vec{c}]}$

If $\vec{r} = \vec{a}$

Then $\lambda(\vec{b} \times \vec{c}) + \mu(\vec{c} \times \vec{a}) + \gamma(\vec{a} \times \vec{b}) = \vec{a}$

$\&(\vec{a} \cdot \vec{a})(\vec{b} \times \vec{c}) + (\vec{b} \cdot \vec{a})(\vec{c} \times \vec{a}) + (\vec{c} \cdot \vec{a})(\vec{a} \times \vec{b}) = \vec{a}[\vec{a}\vec{b}\vec{c}]$

Q.5 Answer:

$\overrightarrow{a}$

Solution:

Since, $\overrightarrow{b}, \overrightarrow{c}$ **and** $\overrightarrow{b} \times \overrightarrow{c}$ **are mutually perpendicular**

vectors, therefore any vector $\vec{a}$ **can be expressed in terms of** $\overrightarrow{b}, \overrightarrow{c}$ **and** $\overrightarrow{b} \times \overrightarrow{c}$

Let $\overrightarrow{a} = x\overrightarrow{b} + y\overrightarrow{c} + z(\overrightarrow{b} \times \overrightarrow{c})$

Taking dot product with $\overrightarrow{b} \times \overrightarrow{c}$ **in eq.** (1)**,**

we get, $\vec{a} \cdot (\vec{b} \times \vec{c}) = 0 + 0 + z|\vec{b} \times \vec{c}|^2$

$\Rightarrow \quad z = \frac{\vec{a} \cdot (\vec{b} \times \vec{c})}{|\vec{b} \times \vec{c}|^2}$

Taking dot product with $\vec{b}$ **in eq. (1), we get** $\vec{a} \cdot \vec{b} = x\vec{b} \cdot \vec{b} + y\vec{c} \cdot \vec{b} + z \cdot 0 = x$

Taking dot product with $\vec{c}$ **in eq. (1), we get**

$\vec{a} \cdot \vec{c} = 0 + y + 0 \Rightarrow y = \vec{a} \cdot \vec{c}$

$\therefore \overrightarrow{a} = (\overrightarrow{a} \cdot \overrightarrow{b})\overrightarrow{b} + (\overrightarrow{a} \cdot \overrightarrow{c})\overrightarrow{c} + \frac{\overrightarrow{a} \cdot (\overrightarrow{b} \times \overrightarrow{c})}{|\overrightarrow{b} \times \overrightarrow{c}|^2}(\overrightarrow{b} \times \overrightarrow{c})$

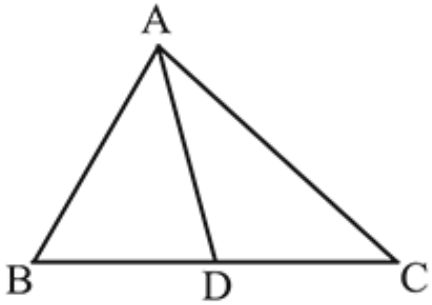
Q.6 Answer:

$\frac{1}{3}(6\hat{i} + 13\hat{j} + 18\hat{k})$

Solution:

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Let O be the origin and the bisector of $\angle A$ meets BC at D. Then $\frac{\mathrm{BD}}{\mathrm{DC}}=\frac{\mathrm{AB}}{\mathrm{AC}}$ and position vector of D is given by



$\overrightarrow{\mathrm{OD}}=\frac{|\overrightarrow{\mathrm{AB}}|\overrightarrow{\mathrm{OC}}+|\overrightarrow{\mathrm{AC}}|\overrightarrow{\mathrm{OB}}}{|\overrightarrow{\mathrm{AB}}|+|\overrightarrow{\mathrm{AC}}|}$

$\overrightarrow{\mathrm{AB}}=2\hat{\mathrm{i}}+3\hat{\mathrm{j}}+4\hat{\mathrm{k}}-4\hat{\mathrm{i}}-7\hat{\mathrm{j}}-8\hat{\mathrm{k}}$

$|\overrightarrow{\mathrm{AB}}|=|-2\mathrm{i}-4\hat{\mathrm{j}}-4\hat{\mathrm{k}}|$

$=\sqrt{(-2)^2+(-4)^2+(-4)^2}=\sqrt{4+16+16}=\sqrt{36}$

$|\overrightarrow{AB}|=6$

$\overrightarrow{\mathrm{AC}}=2\hat{i}+5\hat{j}+7\hat{k}-4\hat{i}-7\hat{j}-8\hat{k}$

$|\overrightarrow{AC}|=|-2i-2\hat{j}-\dot{k}|=\sqrt{(-2)^2+(-2)^2+(-1)^2}$

$=\sqrt{4+4+1}=\sqrt{9}$

$|\overrightarrow{AC}|=3$

$\therefore$ pv, of $\overrightarrow{\mathrm{D}}$

$=\frac{6(2\hat{i}+5\hat{j}+7\hat{k})+3(2\hat{i}+3\hat{j}+4\hat{k})}{6+3}$

$=\frac{12\hat{i}+30\hat{j}+42\hat{k}+6\hat{i}+9\hat{j}+12\hat{k}}{9}$

$=\frac{18\hat{i}+39\hat{j}+54\hat{k}}{9}=\frac{3(6\hat{i}+13\hat{j}+18\hat{k})}{9}$

$=\frac{6\hat{i}+13\hat{j}+18\hat{k})}{3}$

Q.7 Answer:

$\alpha=1,\beta=1$

Solution:

$\because \vec{a}$ lies in the plane of $\vec{b}$ and $\vec{c}$

$\therefore \quad \vec{a}=\vec{b}+\lambda\vec{c}$

$\Rightarrow \alpha\hat{i}+2\hat{j}+\beta\hat{k}=\hat{i}+\hat{j}+\lambda(\hat{j}+\hat{k})$

$\Rightarrow \alpha=1,2=1+\lambda,\beta=\lambda\Rightarrow\alpha=1,\beta=1$

Q.8 Answer:

5

Solution:

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If two vectors are perpedicular to each other then their dot product is equal to 0.

Here, vector $\vec{A}+t\vec{B}$ is perpendicular to $\vec{C}$.

Therefore, $\vec{A}+t\vec{B}\cdot\vec{C}=0$

Now, $\vec{A}+t\vec{B}=\hat{i}+2\hat{j}+3\hat{k}+t\text{-}\hat{i}+2\hat{j}+\hat{k}$

$=1\text{-}t\hat{i}+2+2t\hat{j}+3+t\hat{k}$

$\Rightarrow\vec{A}+t\vec{B}\cdot\vec{C}=1\text{-}t\hat{i}+2+2t\hat{j}+3+t\hat{k}\cdot3\hat{i}+\hat{j}=0$

$\Rightarrow 31\text{-}t+2+2t=0\Rightarrow t=5$

Q.9 Answer:

$\frac{2}{3}$

Solution:

Since, u and v are collinear, therefore $ku+v=0$

$\Rightarrow[k(\alpha-2)+2+3\alpha]\vec{a}+(k-3)\vec{b}=0$

Since $\vec{a}$ and $\vec{b}$ are non-collinear, then for some constant m and n,

$m\vec{a}+n\vec{b}=0\Rightarrow m=0,n=0$

Hence from equation (i) $k-3=0\Rightarrow k=3$

And $k(\alpha-2)+2+3\alpha=0$

$\Rightarrow 3(\alpha-2)+2+3\alpha=0\Rightarrow\alpha=\frac{2}{3}$

Q.10 Answer:

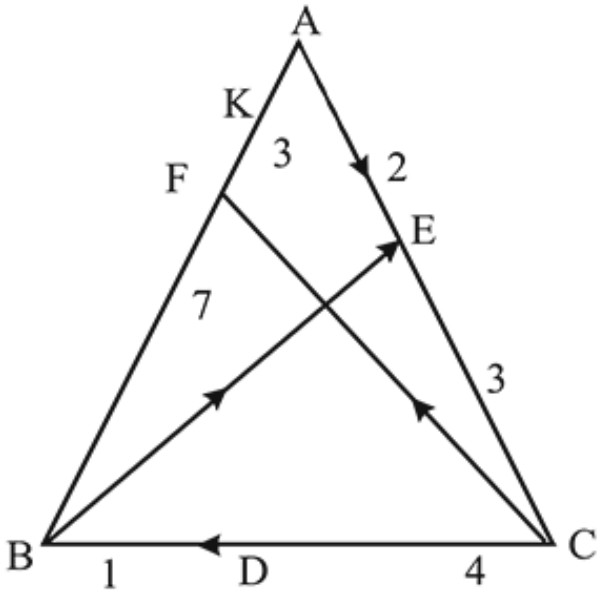
$2:5$

Solution:

Let $\overrightarrow{AB}=a,\overrightarrow{AC}=b$

So, $\overrightarrow{AD}=\frac{4a+b}{5},\overrightarrow{AE}=\frac{2b}{5}$

$\overrightarrow{AF}=\frac{3a}{10}$**, and** $\overrightarrow{AK}=\frac{a}{4}$



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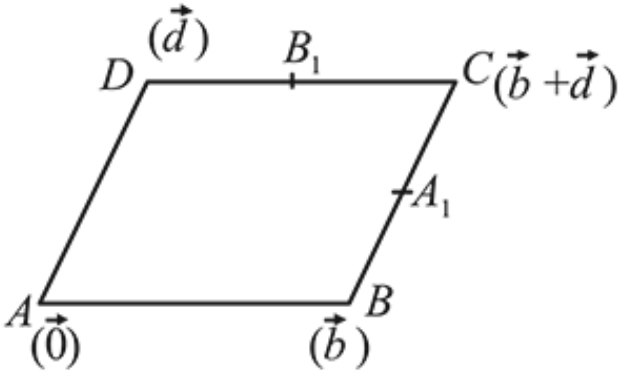
$$\frac{\overrightarrow{AD}+\overrightarrow{BE}+\overrightarrow{CF}}{\overrightarrow{CK}} = \frac{\frac{b+4a}{5}+\frac{2b-5a}{5}+\frac{3a-10b}{10}}{\frac{a-4b}{4}}$$
$$= \frac{6b-2a+3a-10b}{10(a-4b)} \times 4 = \frac{2}{5}.$$

Q.11 Answer:

$\frac{3}{2}$

Solution:

Let P.V. of

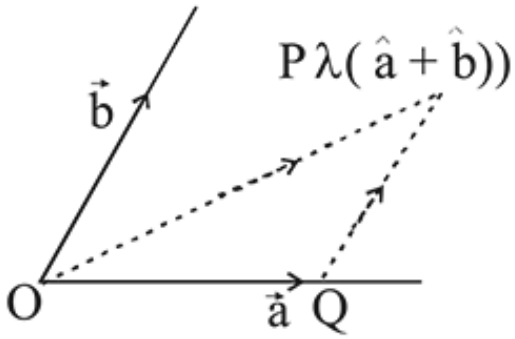


A, B **and** D **be** $\overrightarrow{0}, \vec{b}$ **and** $\vec{d}$**, respectively. Then P.V. of** $C, \vec{c} = \vec{b} + \vec{d}$.
Also P.V. of $A_1 = \vec{b} + \frac{\vec{d}}{2}$
and P.V. of $B_1 = \vec{d} + \frac{\vec{b}}{2} \Rightarrow \overrightarrow{AA_1} + \overrightarrow{AB_1} = \frac{3}{2}(\vec{b} + \vec{d}) = \frac{3}{2}\overrightarrow{AC}$

Q.12 Answer:

$\hat{\mathrm{i}} - 7\hat{\mathrm{j}} + 2\hat{\mathrm{k}}$

Solution:



$\mathrm{OQ} = \mathrm{PQ}$
$\mathrm{OQ} = \mathrm{PQ} = \lambda(\text{say})$
$\overrightarrow{\mathrm{OP}} = \overrightarrow{\mathrm{OQ}} + \overrightarrow{\mathrm{QP}}; \overrightarrow{\mathrm{c}} = \lambda\hat{\mathrm{a}} + \lambda\hat{\mathrm{b}}$
Let $\hat{a}$ **and** $\hat{b}$ **be unit**
vectors along $\vec{a}$ **and** $\vec{b}$ **respectively, then** $\hat{a} = \frac{1}{9}(7\hat{i} - 4\hat{j} - 4\hat{k})$ **and**
$\hat{\mathrm{b}} = \frac{1}{3}(-2\hat{\mathrm{i}} - \hat{\mathrm{j}} + 2\hat{\mathrm{k}})$
The required vector

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$\overrightarrow{c} = \lambda(\hat{a} + \hat{b})$, **where** λ **is a scalar**

$= \lambda\left(\frac{1}{9}\hat{i} - \frac{7}{9}\hat{j} + \frac{2}{9}\hat{k}\right)$

$|\overrightarrow{c}|^2 = \lambda^2\left(\frac{1}{81} + \frac{49}{81} + \frac{4}{81}\right) = \frac{54}{81}\lambda^2$

$\Rightarrow (3\sqrt{6})^2 = \frac{54}{81}\lambda^2$

$\Rightarrow \lambda^2 = 81 \Rightarrow \lambda = \pm 9$. **Hence,** $\vec{c} = (\hat{i} - 7\hat{j} + 2\hat{k})$

Q.13 Answer:

Geometric mean of α and β

Solution:

If three vectors are co-planar. $\Rightarrow \begin{vmatrix} \alpha & \alpha & \gamma \\ 1 & 0 & 1 \\ \gamma & \gamma & \beta \end{vmatrix} = 0$

$\Rightarrow \quad \alpha[0 - \gamma] - \alpha[\beta - \gamma] + \gamma[\gamma - 0] = 0$

$\Rightarrow -\alpha\gamma - \alpha\beta + \alpha\gamma + \gamma^2 = 0$

$\Rightarrow \quad \gamma^2 = \alpha\beta$

$\Rightarrow$ **So** α, γ, β **are in G.P.**

Q.14 Answer:

0

Solution:

Since, $a + 2b$ **is collinear with c**

$a + 2b = mc \ldots (\mathrm{i})$

Since $b + 3c$ **is collinear with** a

$\mathrm{b} + 3\mathrm{c} = \mathrm{na} \ldots\ldots (\mathrm{ii})$

Myltiplying Eq(ii) by 2 and subtracting Eq.(ii) from Eq.(i), we get $a - bc = mc - 2na$

On comparing, we get

$\mathrm{m} = -6; \mathrm{n} = -\frac{1}{2}$

From Eq.(i), we have

$a + 2b = -6c$

$\Rightarrow a + b + 6c = 0$

Q.15 Answer:

form an equilateral triangle

Solution:

Let $\mathrm{OP} = \alpha\mathrm{i} + \beta\mathrm{j} + \gamma\mathrm{k}$,

$\mathrm{OQ} = \beta\mathrm{i} + \gamma\mathrm{j} + \alpha\mathrm{k}$

and $\mathrm{OR} = \gamma\mathrm{i} + \alpha\mathrm{j} + \beta\mathrm{k}$

Now, $\mathrm{PQ} = \mathrm{OQ} - \mathrm{OP}$

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$= (\beta - \alpha)\mathrm{i} + (\gamma - \beta)\mathrm{j} + (\alpha - \gamma)\mathrm{k}$

Also, $\mathrm{PR} = \mathrm{OR} - \mathrm{OP}$

$= (\gamma - \alpha)\mathrm{i} + (\alpha - \beta)\mathrm{j} + (\beta - \gamma)\mathrm{k}$

Again, $\mathrm{QR} = \mathrm{OR} - \mathrm{OQ}$

$= (\gamma - \beta)\mathrm{i} + (\alpha - \gamma)\mathrm{j} + (\beta - \alpha)\mathrm{k}$

Thus, $|\mathrm{PQ}| = |\mathrm{QR}| = |\mathrm{PR}|$

$= \sqrt{(\beta - \alpha)^2 + (\gamma - \beta)^2 + (\alpha - \gamma)^2}$

Therefore, $\triangle \mathrm{PQR}$ **is an equilateral triangle.**

Q.16 Answer:

$\alpha = \pm 1, \beta = 1$

Solution:

Since $\vec{a}, \vec{b}$ **and** $\vec{c}$ **are coplanar therefore** $\begin{vmatrix} 1 & 1 & 1 \\ 4 & 3 & 4 \\ 1 & \alpha & \beta \end{vmatrix} = 0 \Rightarrow \beta = 1; |\vec{c}| = \sqrt{1 + \alpha^2 + \beta^2} = \sqrt{3}$

$\Rightarrow \alpha^2 + \beta^2 = 2 \Rightarrow \alpha^2 = 1 \therefore \alpha = \pm 1$

Q.17 Answer:

linearly independent vectors

Solution:

Since, $\vec{a}$ **and** $\vec{b}$ **are non-collinear, so** $\vec{a} + \vec{b}$ **and** $\vec{a}$ - $\vec{b}$ **will also be non-collinear.**

Hence, $\vec{a} + \vec{b}$ **and** $\vec{a}$ - $\vec{b}$ **are linearly independent vectors**

Also, if we assume $\vec{a}, \vec{b}$ **as two adjacent sides of a parallelogram then** $\vec{a} + \vec{b}$ & $\vec{a}$ - $\vec{b}$ **represents the vectors of corresponding diagonals of parallelogram.**

So, these two are linearly independent.

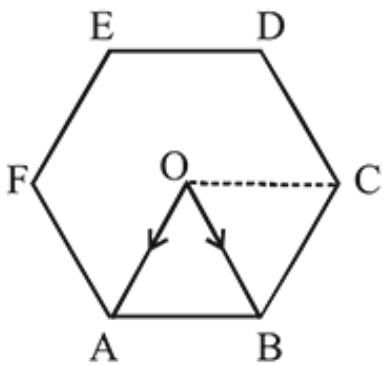
Q.18 Answer:

$-\hat{\mathrm{i}} + \hat{\mathrm{j}} - 2\hat{\mathrm{k}}$

Solution:

$\overrightarrow{\mathrm{OA}} = \hat{\mathrm{i}} - \hat{\mathrm{j}} + 2\hat{\mathrm{k}}, \overrightarrow{\mathrm{OB}} = 2\hat{\mathrm{i}} + \hat{\mathrm{j}} - \hat{\mathrm{k}}$

$\therefore \overrightarrow{\mathrm{OC}} = \overrightarrow{\mathrm{AB}} = \overrightarrow{\mathrm{OB}} - \overrightarrow{\mathrm{OA}} = \hat{\mathrm{i}} + 2\hat{\mathrm{j}} - 3\hat{\mathrm{k}}$



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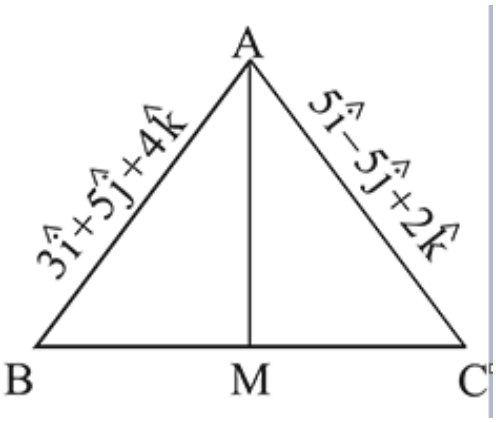
$\therefore \overrightarrow{BC} = \overrightarrow{OC} - \overrightarrow{OB} = (\hat{i} + 2\hat{j} - 3\hat{k}) - (2\hat{i} + \hat{j} - \hat{k})$
$= -\hat{i} + \hat{j} - 2\hat{k}$

Q.19 Answer:

5 units

Solution:

Let the given vectors be $\overrightarrow{AB} = 3\hat{i} + 5\hat{j} + 4\hat{k}$
and $\overrightarrow{AC} = 5\hat{i} - 5\hat{j} + 2\hat{k}$



Let AM be the median through $A \therefore \quad \overrightarrow{AM} = \frac{1}{2}(\overrightarrow{AB} + \overrightarrow{AC})$
$= \frac{1}{2}[(3\hat{i} + 5\hat{j} + 4\hat{k}) + (5\hat{i} - 5\hat{j} + 2\hat{k})]$
$= \frac{1}{2}(8\hat{i} + 6\hat{k}) = (4\hat{i} + 3\hat{k})$
$\therefore$ **Length of the median** $AM = \sqrt{4^2 + 3^2}$
$=$ **5 units**

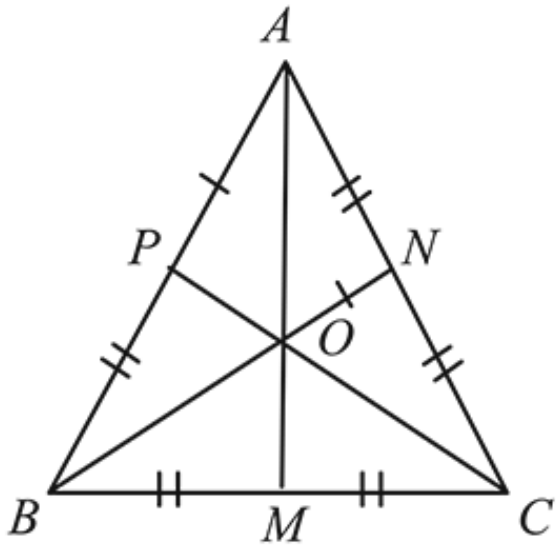
Q.20 Answer:

$\vec{a} + \vec{b} + \vec{c} = \overrightarrow{0}$

Solution:

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Position vectors of vertices A, B and C are $\vec{a}, \vec{b}$ and $\vec{c}$.



$\because$ triangle is equilateral.

$\therefore$ Centroid and orthocenter will coincide. Centroid $\equiv$ orthocenter position vector

$$= \frac{1}{3}(\vec{a} + \vec{b} + \vec{c})$$

$\because$ given in question orthocenter is at origin. Hence $\frac{1}{3}(\vec{a} + \vec{b} + \vec{c}) = 0$

$\vec{a} + \vec{b} + \vec{c} = 0$

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